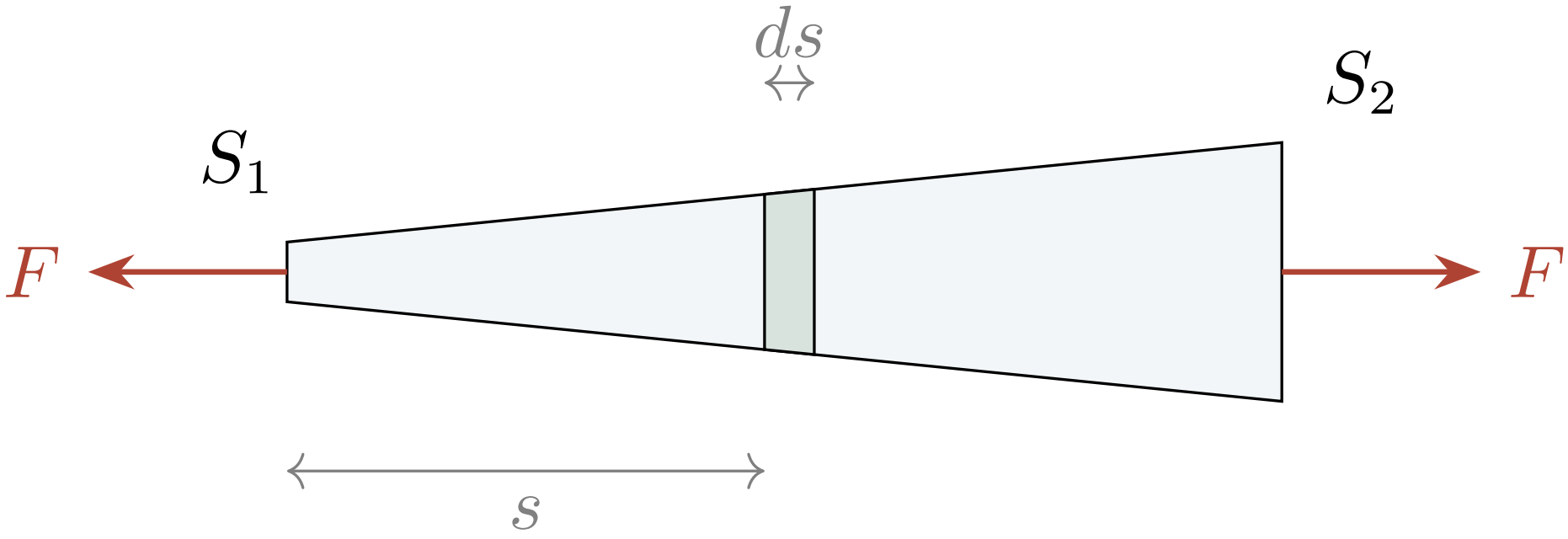

$$S(s) = S_1 + (S_2 - S_1)s/l_0$$

$$d(\Delta l) = F\,ds/[ES(s)]$$